Metro Ticket Generating System

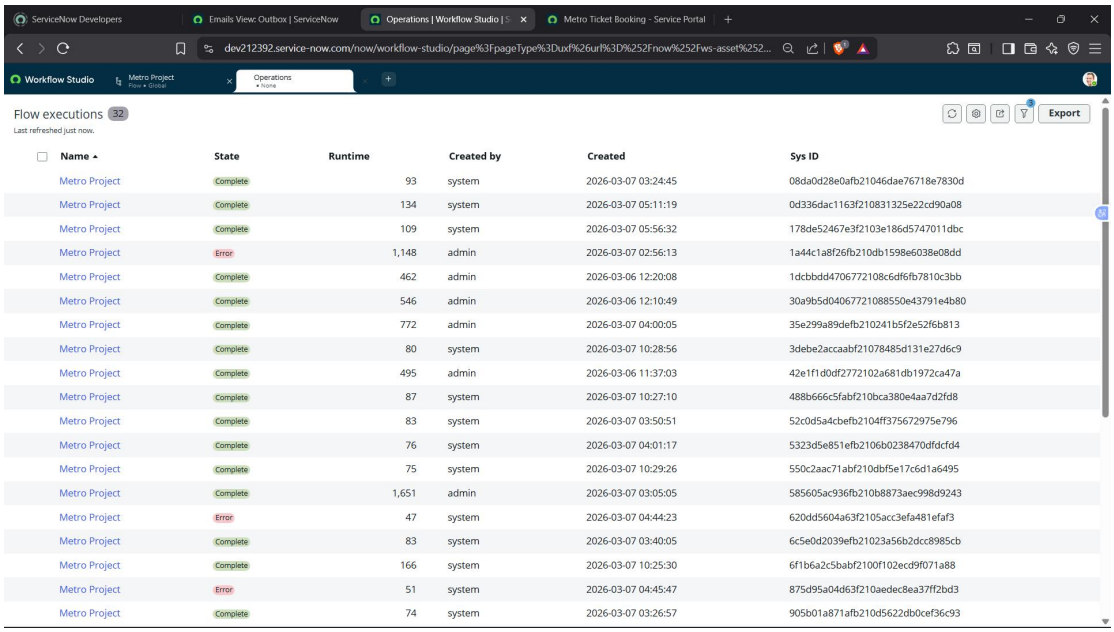
Phase 5-Deployment, Documentation and Final Presentation

1. Troubleshooting

During the development of the Metro Ticket Generating System, several issues were encountered and resolved through systematic debugging and testing. Troubleshooting helped ensure that the application worked smoothly across the Service Portal and backend processes.

Key problems addressed included QR code generation errors, incorrect fare calculations, and variable mapping issues between catalog items and the custom database table. Flow Designer execution logs were used to trace workflow steps and verify that all automation paths executed correctly.

Additionally, form behavior testing was performed to confirm that UI policies and catalog client scripts triggered correctly based on user inputs.



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Metro Project	Error	1,148	admin	2026-03-07 02:56:13	1a44c1a8f26fb210db1598e6038e08dd
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Metro Project	Complete	166	system	2026-03-07 10:25:30	6f1b6a2c5babf2100f102ecd9f071a88
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Metro Project	Complete	74	system	2026-03-07 03:26:57	905b01a871afb210d5622db0cef36c93

2. Adherence to Timelines

The project was executed according to a structured development timeline to ensure that each milestone was completed efficiently. The implementation followed the defined project roadmap covering catalog development, automation logic, user interface configuration, and final testing.

Each phase of the project was completed in an organized sequence including catalog creation, form setup, workflow automation, and system validation. Regular testing ensured that all components were functioning correctly before moving to the next stage.

Following this timeline helped maintain project consistency and ensured timely completion of the Metro Ticket System.

User submits ticket booking form

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Flow Designer Triggered

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Retrieve Catalog Variables

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Create Metro Ticket Record

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Generate Ticket Number

↓

Generate QR Code

↓

Update Ticket Status

↓

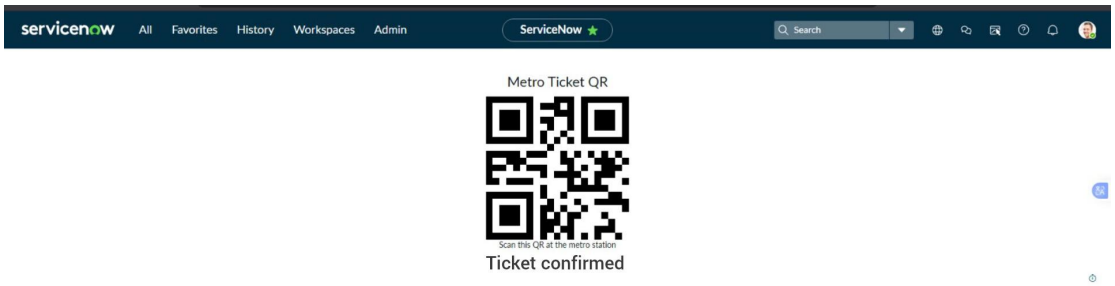
Send Email Notification

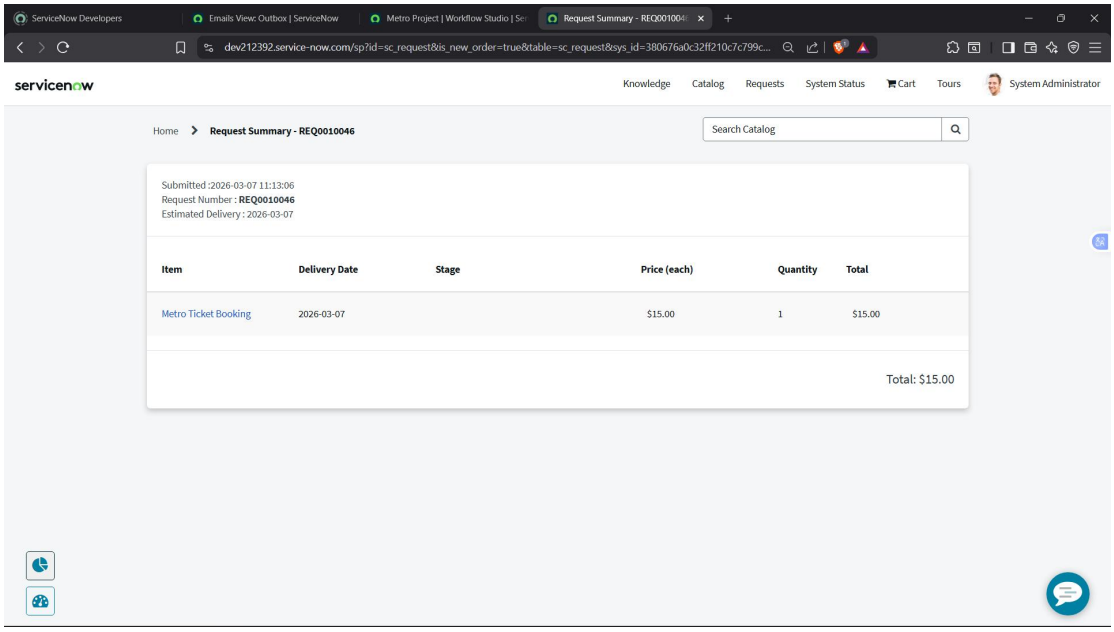
3. Innovation

The project introduces an innovative approach to metro ticket generation by combining ServiceNow automation with QR-based digital ticketing. The system automates ticket booking, fare calculation, data storage, and QR code generation using minimal custom scripting.

Flow Designer was used to implement end-to-end automation for processing ticket requests, while catalog client scripts and UI policies enabled dynamic form behavior. This approach reduces manual intervention and improves system maintainability.

The QR-based ticket allows passengers to quickly validate their travel at metro stations, making the solution efficient and scalable.



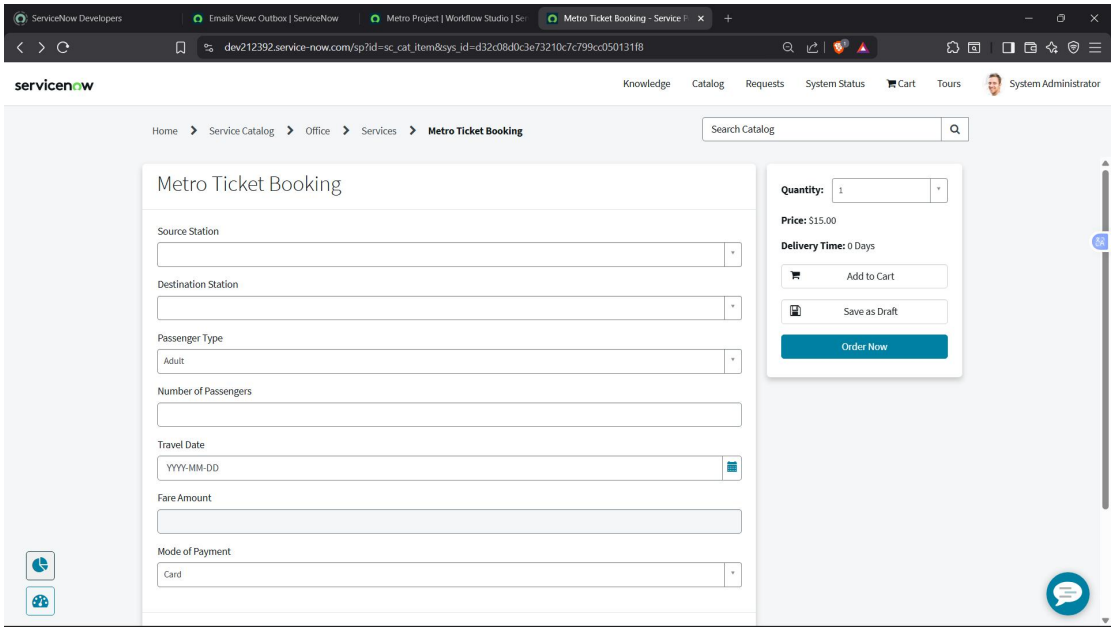


4. Document Functional Overview

The functional overview documentation explains how the Metro Ticket Generating System operates from a user perspective. It describes the booking workflow, form variables, ticket generation process, and QR code validation.

The system allows users to select source and destination stations, choose the number of passengers, and generate a digital metro ticket. Once submitted, the request is processed automatically and the ticket data is stored in the system database.

This documentation helps stakeholders understand the system functionality and overall workflow.

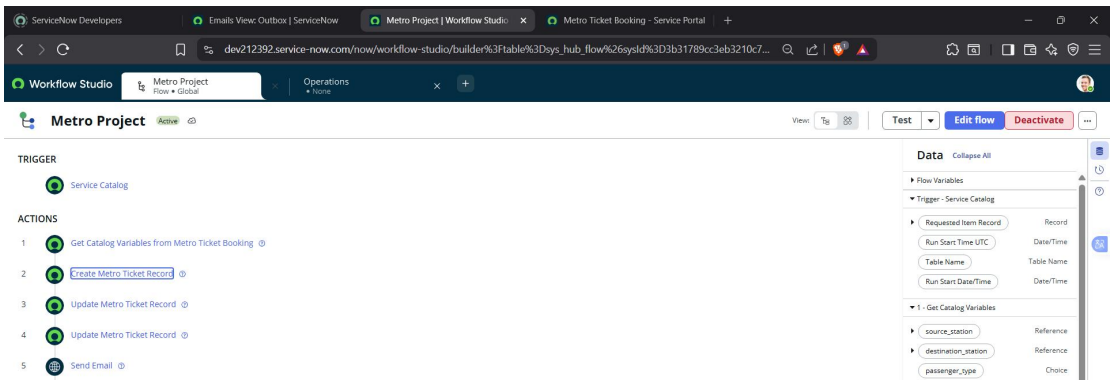


5. Document Technical Blueprint

The technical blueprint explains the architecture and backend components used to build the Metro Ticket System in ServiceNow.

It includes details about catalog item configuration, Flow Designer automation, variable mapping, custom database table structure, and QR code generation logic. The documentation also describes how user inputs are captured and processed within the system.

This technical documentation ensures that future developers or administrators can easily understand and maintain the system.

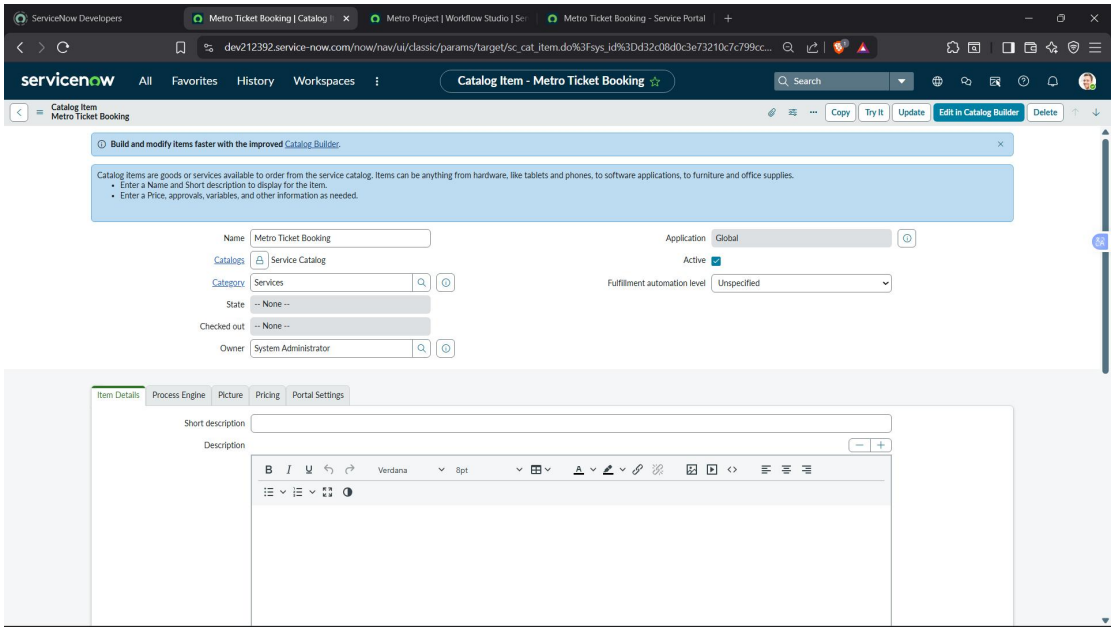


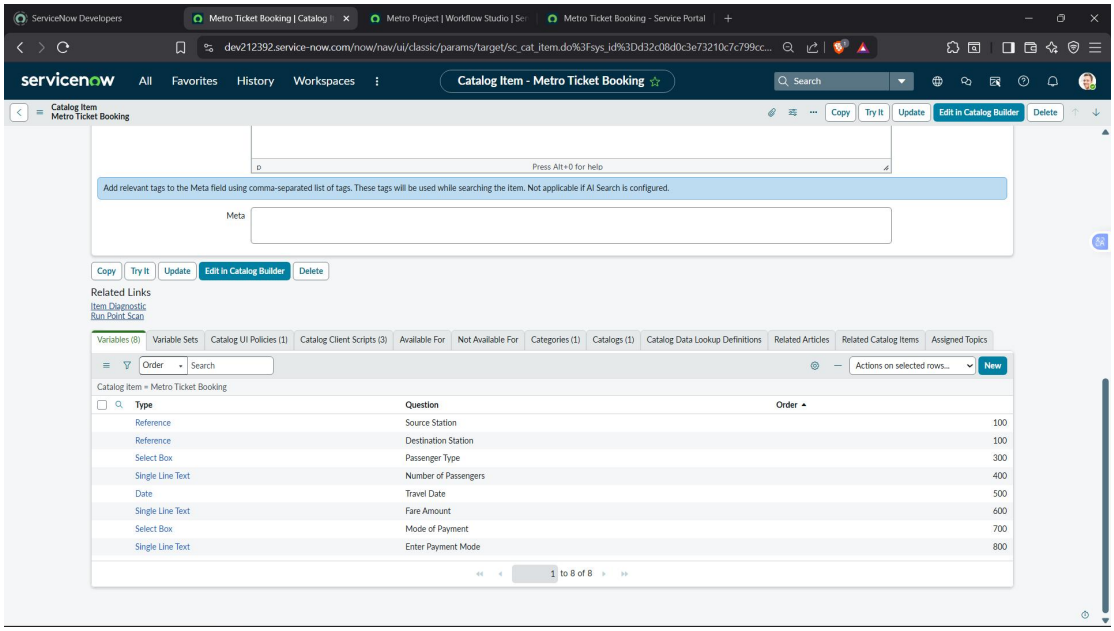
6. Document Setup Manual

The setup manual provides a step-by-step guide for recreating the Metro Ticket Generating System in a new ServiceNow instance. It explains how to configure catalog items, variables, automation workflows, and access control settings.

The manual includes instructions for creating the custom database table, configuring Flow Designer processes, and implementing QR code generation logic. Screenshots and clear instructions are included to simplify deployment.

This setup guide ensures that the system can be replicated easily for testing or production environments.





7. Project Demo Video Planning

The demo video demonstrates the complete working of the Metro Ticket Generating System from ticket booking to QR code validation.

The demonstration includes the Service Portal interface where a user selects stations and submits a ticket request. The system then processes the request, stores ticket details in the database, and generates a QR code for metro station verification.

The video walkthrough highlights both the user experience and the backend automation processes.

8. Visual Demonstration

The visual demonstration section provides screenshots that explain the system workflow step by step. These visuals help illustrate the user interface, backend automation, and QR ticket generation.

Captured visuals include the catalog form interface, dynamic form behavior, Flow Designer execution paths, and the generated QR code. Each screenshot highlights a specific stage of the system process.

This visual documentation makes the project easier to understand and review.

9. Explanation Clarity

The project documentation is structured to clearly explain each step of the Metro Ticket System implementation. Each section follows a logical sequence describing the purpose, implementation, and outcome of the feature.

Clear headings, screenshots, and descriptive explanations are used to make the documentation easy to understand for reviewers and developers. The documentation also includes explanations for automation logic and data flow.

This structured explanation improves readability and supports knowledge transfer.

10. Realism & Quality

The system was designed to simulate a real-world metro ticket booking process similar to those used in modern urban transportation systems.

Realistic scenarios were implemented including fare calculation based on stations, passenger input handling, ticket generation, and QR code validation. Sample data and practical test cases were used to verify system accuracy.

This ensures that the project reflects a practical and real-world ticketing solution.

11. Scalability & Future Plan

The Metro Ticket Generating System is designed with scalability in mind, allowing future enhancements to be added easily. Potential improvements include integration with payment gateways, SMS or email ticket delivery, and mobile application connectivity.

Additional analytics dashboards could also be implemented to track ticket usage, travel patterns, and system performance. These enhancements would make the system more robust and suitable for real-world metro operations.

The modular design ensures that new features can be added without affecting the existing system.